PROBLEM NS6

Sphere with Radial Suction

Original statement · NS6.pdf

Spherical symmetry | Suction | Trajectory and time | Convection-conduction

OFFICIAL PROBLEM STATEMENT

A sphere of radius a is immersed in a liquid of density ρ, viscosity μ, thermal conductivity k, and specific heat c. The liquid is sucked with radial velocity $v_r = -V_o$ through the porous surface of the sphere. For the buoyancy-free, steady sphericosymmetrical motion induced in the surrounding liquid,

1. Obtain the velocity $v_r(r)$

2. Compute the trajectory of a fluid particle initially located at $r = r_i > a$, as well as the time t_i at which the fluid particle is sucked by the sphere.

3. Determine the pressure distribution $p - p_\infty$, where p_∞ is the pressure value far from the sphere.

4. Calculate the temperature distribution in the liquid $T(r)$ assuming that the temperature of the sphere is T_s, smaller then the value T_∞ found in the surrounding ambient atmosphere.

5. Obtain the amount of heat transfered to the sphere per unit time.

Transcribed as printed: "sphericosymmetrical", "then" (for "than") and "transfered" are spelling errors of the sheet. The sheet carries no printed figure.

Source: NS6.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41 and energy Eq. 4.42 in spherical coordinates, viscous dissipation ϕ_v in spherical coordinates.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, spherically symmetric flow of a perfect liquid (constant ρ, μ, k, c): $\bar{v} = v_r(r)\,\bar{e}_r$, $v_\theta = v_\phi = 0, \partial/\partial\theta = \partial/\partial\phi = \partial/\partial t = 0$.
- Buoyancy-free: the mass force is neglected. No chemical reaction and no radiation. The temperature depends only on r.

- Suction: the porous sphere takes liquid in at the rate $4\pi a^2 V_o$, with $V_o > 0$. Far from the sphere the liquid is at rest at p_∞ and T_∞.
- Data: $[\rho,\ \mu,\ k,\ c,\ a,\ V_o,\ T_s,\ T_\infty,\ p_\infty,\ r_i]$ with $T_s < T_\infty$. Unknowns: $v_r(r), r(t), t_i$, $p(r), T(r)$ and the heat rate to the sphere.
- Dimensionless numbers used: the suction Péclet number $\mathrm{Pe} = \rho c a V_o / k$ and the length $\beta = \rho c a^2 V_o / k = a\,\mathrm{Pe}$.

Symbol	Meaning	SI unit
a	sphere radius	m
V_o	suction velocity at the sphere surface	m/s
r_i, t_i	initial position of the particle, time to reach the sphere	m, s
T_s, T_∞	sphere and ambient temperatures	K
Pe, β	suction Péclet number, length $\beta = a\,\mathrm{Pe}$	1, m
$\Delta T_d = 12\mu V_o/(\rho c a)$	temperature scale of the viscous heating	K

PHASE 2

Frames, Field Form and Boundary Conditions

- Spherical frame (r, θ, ϕ) centred on the sphere; $\bar{e}_r$ points away from the centre; the liquid occupies $r > a$.
- Field form: $\bar{v} = v_r(r)\,\bar{e}_r, p = p(r), T = T(r)$.

Location	Condition	Origin
$r = a$	$v_r = -V_o$	suction through the porous surface
$r \to \infty$	$p \to p_\infty$	statement
$r = a$	$T = T_s$	temperature of the sphere
$r \to \infty$	$T \to T_\infty$	ambient temperature

PHASE 3

Step-by-Step Derivation

3.1 Continuity and the radial velocity

$$\frac{1}{r^2}\frac{d(r^2 v_r)}{dr} = 0 \implies r^2 v_r = \text{const} = -a^2 V_o \tag{1}$$

where the constant is fixed with $v_r(a) = -V_o$:

$$\boxed{v_r(r) = -\frac{a^2 V_o}{r^2}} \tag{2}$$

3.2 Trajectory and suction time

A fluid particle moves with the local velocity: $dr/dt = v_r = -a^2 V_o / r^2$. Separating variables, $r^2\, dr = -a^2 V_o\, dt$. Integrating from r_i at $t = 0$:

$$\frac{r^3 - r_i^3}{3} = -a^2 V_o\, t \implies \boxed{r(t) = \left(r_i^3 - 3a^2 V_o\, t\right)^{1/3}} \tag{3}$$

The particle is sucked when $r = a$: $a^3 = r_i^3 - 3a^2 V_o\, t_i$, so

$$\boxed{t_i = \frac{r_i^3 - a^3}{3\, a^2 V_o}} \tag{4}$$

3.3 Pressure distribution

In the r component of (4.41) the viscous term $\mu\,\partial\left[r^{-2}\partial(r^2 v_r)/\partial r\right]/\partial r$ vanishes because $r^2 v_r$ is constant (1): the flow is irrotational. The rest is the steady Euler equation:

$$\rho\, v_r \frac{dv_r}{dr} = -\frac{dp}{dr} \;\Longrightarrow\; p + \frac{\rho v_r^2}{2} = \text{const} = p_\infty \tag{5}$$

(the constant is evaluated at infinity, where $v_r \to 0$). Substituting (2):

$$\boxed{p - p_\infty = -\frac{\rho\, a^4 V_o^2}{2\, r^4}} \tag{6}$$

The pressure is below the ambient value, the minimum being at the sphere surface, where the liquid has the highest speed.

3.4 Temperature distribution

With $v_\theta = v_\phi = 0$ the energy equation (4.42) in spherical coordinates is

$$\rho c\, v_r \frac{dT}{dr} = \phi_v + \frac{k}{r^2}\frac{d}{dr}\left(r^2 \frac{dT}{dr}\right) \tag{7}$$

The viscous dissipation is $\phi_v = 2\mu\left[(dv_r/dr)^2 + 2(v_r/r)^2\right]$ with $dv_r/dr = 2a^2 V_o / r^3$ and $v_r/r = -a^2 V_o / r^3$:

$$\phi_v = 2\mu\left[\frac{4a^4 V_o^2}{r^6} + \frac{2a^4 V_o^2}{r^6}\right] = \frac{12\mu\, a^4 V_o^2}{r^6} \tag{8}$$

Substituting v_r and ϕ_v in (7), dividing by k and multiplying by r^2:

$$\frac{d}{dr}\left(r^2\frac{dT}{dr}\right) + \beta\frac{dT}{dr} = -\frac{\Gamma}{r^4}, \qquad \beta = \frac{\rho c a^2 V_o}{k}, \quad \Gamma = \frac{12\mu a^4 V_o^2}{k} \tag{9}$$

Homogeneous solution. With $y = T'$: $r^2 y' + (2r + \beta) y = 0$; the integrating factor is $\exp \int (2/r + \beta/r^2)\, dr = r^2 e^{-\beta/r}$, so $y = D\, e^{\beta/r}/r^2$, and since $\int e^{\beta/r} r^{-2} dr = -e^{\beta/r}/\beta$:

$$T_h = B_2 + B_1\, e^{\beta/r} \tag{10}$$

Particular solution. Work with $y = T_p'$ in the first-order form $r^2 y' + (2r + \beta) y = -\Gamma / r^4$. The right-hand side is a power of $1/r$, so try a sum of powers, $y = -\Gamma \left(\frac{c_4}{r^4} + \frac{c_3}{r^3} + \frac{c_2}{r^2} \right)$, and match powers one at a time. The three terms of the equation are $r^2 y' = \Gamma \left(\frac{4c_4}{r^3} + \frac{3c_3}{r^2} + \frac{2c_2}{r} \right)$, $2ry = -\Gamma \left(\frac{2c_4}{r^3} + \frac{2c_3}{r^2} + \frac{2c_2}{r} \right)$ and $\beta y = -\Gamma \left(\frac{\beta c_4}{r^4} + \frac{\beta c_3}{r^3} + \frac{\beta c_2}{r^2} \right)$. Their sum is $\Gamma \left[-\frac{\beta c_4}{r^4} + \frac{2c_4 - \beta c_3}{r^3} + \frac{c_3 - \beta c_2}{r^2} \right]$, which must equal $-\Gamma / r^4$: $c_4 = 1/\beta$, $c_3 = 2c_4/\beta = 2/\beta^2$ and $c_2 = c_3/\beta = 2/\beta^3$. Hence

$$y = T_p' = -\Gamma \left(\frac{1}{\beta r^4} + \frac{2}{\beta^2 r^3} + \frac{2}{\beta^3 r^2} \right)$$

and integrating y with respect to r (the antiderivative of r^{-n} is $-r^{1-n}/(n-1)$):

$$T_p(r) = \Gamma \left(\frac{1}{3\beta r^3} + \frac{1}{\beta^2 r^2} + \frac{2}{\beta^3 r} \right) = \Delta T_d \left[\frac{1}{3} \left(\frac{a}{r} \right)^3 + \frac{1}{\mathrm{Pe}} \left(\frac{a}{r} \right)^2 + \frac{2}{\mathrm{Pe}^2} \frac{a}{r} \right] \tag{11}$$

with $\Delta T_d = 12 \mu V_o / (\rho c a)$ (use $\Gamma/\beta = 12 \mu a^2 V_o / (\rho c)$ and $\beta = a\,\mathrm{Pe}$). The general solution is $T = B_2 + B_1 e^{\beta/r} + T_p(r)$.

Boundary conditions. $T(\infty) = T_\infty$: $e^{\beta/r} \to 1$ and $T_p \to 0$, so $B_2 + B_1 = T_\infty$. $T(a) = T_s$: $B_2 + B_1 e^{\mathrm{Pe}} + T_p(a) = T_s$, because $\beta/a = \mathrm{Pe}$. Subtracting,

$$B_1 = -\frac{T_\infty - T_s + T_p(a)}{e^{\mathrm{Pe}} - 1}, \qquad T_p(a) = \Delta T_d \left(\frac{1}{3} + \frac{1}{\mathrm{Pe}} + \frac{2}{\mathrm{Pe}^2} \right)$$

$$\boxed{T(r) = T_\infty + T_p(r) - [T_\infty - T_s + T_p(a)]\,\frac{e^{\mathrm{Pe}\,a/r} - 1}{e^{\mathrm{Pe}} - 1}} \qquad (12)$$

Discrepancy with the official solution

The handwritten solution gets the particular solution $12\mu V_o/(\rho c\, r)$, as read from the handwriting. Its ODE has the dissipation term as $12\mu V_o^2 a^2/(k\, r^2)$, but multiplying ϕ_v/k of (8) by r^2 gives $12\mu a^4 V_o^2/(k\, r^4)$, as in (9). The term A/r cannot satisfy (9), whose right-hand side decays as r^{-4}: the check above shows that (11) does. The computed result (12) is kept; with $\Delta T_d \to 0$ (negligible dissipation) it reduces to the sheet's result $T = T_\infty + (T_s - T_\infty)\dfrac{e^{\mathrm{Pe}\,a/r} - 1}{e^{\mathrm{Pe}} - 1}$.

3.5 Heat transferred to the sphere

The heat that flows from the liquid into the sphere per unit time is $4\pi a^2\, \bar{q}\cdot\bar{n}$ with $\bar{n} = -\bar{e}_r$ (outward from the liquid) and $\bar{q} = -k\nabla T$, that is $\dot{Q}_s = 4\pi a^2 k\, dT/dr\big|_a$. From (12),

$$\left.\frac{dT}{dr}\right|_a = T_p'(a) + \frac{\mathrm{Pe}}{a}\,[T_\infty - T_s + T_p(a)]\,\frac{e^{\mathrm{Pe}}}{e^{\mathrm{Pe}} - 1}, \qquad T_p'(a) = -\frac{\Delta T_d}{a}\left(1 + \frac{2}{\mathrm{Pe}} + \frac{2}{\mathrm{Pe}^2}\right) \qquad (13)$$

Using $4\pi a\, k\,\mathrm{Pe} = 4\pi\rho c\, a^2 V_o$ and $e^{\mathrm{Pe}}/(e^{\mathrm{Pe}} - 1) = 1/(1 - e^{-\mathrm{Pe}})$:

$$\boxed{\dot{Q}_s = \frac{4\pi\rho c\, a^2 V_o\,[T_\infty - T_s + T_p(a)]}{1 - e^{-\mathrm{Pe}}} - 4\pi a\, k\, \Delta T_d \left(1 + \frac{2}{\mathrm{Pe}} + \frac{2}{\mathrm{Pe}^2}\right)} \qquad (14)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** The suction brings ambient liquid at T_∞ towards the cold sphere, which compresses the thermal layer against the surface: $\beta = a\,\mathrm{Pe}$ is the length over which convection and conduction compete. The temperature field (12) has no solution that is regular at infinity for blowing ($V_o < 0$), where the exponent would change sign.

- **Limits.** For negligible dissipation ($\Delta T_d \to 0$): $\dot{Q}_s = 4\pi\rho c a^2 V_o\,(T_\infty - T_s)/(1 - e^{-\mathrm{Pe}})$. For $\mathrm{Pe} \gg 1$ this is $\dot{m}\,c\,(T_\infty - T_s)$ with $\dot{m} = 4\pi\rho a^2 V_o$ the sucked mass flow: the sphere only has to cool the liquid it takes in. For $\mathrm{Pe} \to 0$ it tends to $4\pi a k\,(T_\infty - T_s)$, pure conduction to a sphere in an unbounded medium.
- **Dimensional check.** $a^2 V_o / r^2$ is a velocity. ρv_r^2 in (6) is a pressure. $\beta = \rho c a^2 V_o / k$ has units $(\mathrm{J}/(\mathrm{m^3 K}))\,\mathrm{m^3/s}/(\mathrm{W}/(\mathrm{m\,K})) = \mathrm{m}$. $\Delta T_d = 12\mu V_o/(\rho c a)$ has units $\mathrm{Pa\,s}\cdot\mathrm{m/s}/[(\mathrm{Pa/K})\,\mathrm{m}] = \mathrm{K}$. $\dot{Q}_s$ in (14): $\rho c a^2 V_o\,\Delta T$ has units $(\mathrm{Pa/K})\,\mathrm{m^3/s}\cdot\mathrm{K} = \mathrm{W}$.
- **Check of (11).** Differentiating the three terms of (11) and substituting in (9) reproduces $-\Gamma/r^4$ term by term, as written in 3.4.